Experimental study of acoustic resonance in a rectangular cavity

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The resonance frequency and sound pressure distribution within a cavity offer highly interesting applications in acoustics, such as the acoustic design of buildings and the design and tuning of musical instruments. This experiment studies the sound pressure distribution characteristics of a rectangular cavity and compares the experimentally measured resonance frequency with the predicted value of the theoretical model. In the experimental procedure, a Bluetooth speaker generated sound at different frequencies inside an enclosed wooden cavity, and a microphone connected to an oscilloscope recorded sound pressure amplitude, which allows identification of resonance frequencies. The measured frequencies, such as 147.0 ± 3.5 Hz and 308.0 ± 4.5 Hz, deviate significantly from the most likely theoretical predictions (143.3 ± 6.0 Hz and 286.7 ± 11.9 Hz, respectively). These discrepancies indicate that the theoretical resonance frequency model systematically underestimates the actual resonance, and because of the cavity wall vibration and sound leakage observed in the experiment, this suggests that we need to consider non-ideal rigid boundary conditions. The results highlight the need to consider realistic boundary conditions when predicting acoustic resonances in practical situations.

I. INTRODUCTION

This experiment explores acoustic resonance in a cavity, and a relevant application is understanding how sound is amplified or diminished in enclosed building spaces. Intuitively, we might expect that in an enclosed environment with a fixed sound source, the sound volume decreases with distance, but due to resonance effects, the actual distribution of sound pressure can be far more complex. Several key concepts are central to this experiment:

- **Stationary waves**: Two waves traveling in opposite directions with the same amplitude then interfere and create a stationary wave [1].
- **Cavity natural resonance frequencies**: Resonance occurs when a given system is driven externally to vibrate at greater amplitude at a certain preferred frequency. This frequency is known as the resonant frequency, which corresponds to the natural frequency of vibration of the object or system. For an acoustical cavity, the resonant frequencies will depend upon the dimensions of the cavity, its geometry, and the speed of sound at the temperature of the air [2].
- **Sound pressure**: The change in pressure caused by vibration when sound waves pass through a medium.

Additionally, the theoretical framework relies on solving the three-dimensional wave equation by using the separation of variables and applying the ideal gas law to determine the speed of sound. More specifically, since air can be approximated as an ideal gas, the speed of sound in air at temperature T can be calculated using the following formula:

- $v_{ideal} = \sqrt{\frac{\gamma k T}{m}}$, where γ is the adiabatic index, T is the temperature and k is the Boltzmann constant.

In this experiment, we used a rectangular wooden cavity as the object, placed a Bluetooth speaker at the center of the bottom of the cavity, and placed a microphone connected to an oscilloscope at the midpoint of the cavity's bottom edge to analyze the sound pressure distribution. We set the speaker's frequency as the independent variable, and measured the corresponding changes in sound pressure amplitude displayed on the oscilloscope as the dependent variable. The measured resonance frequency is compared with the predicted resonance frequency of the possible resonance mode. We used the 68% confidence interval in statistics to determine the uncertainty of the measured values and the theoretical values, and quantified the validity of the theoretical prediction by calculating the ratio of the area of overlap of the Gaussian distributions of the two to the area of the measured value distribution. **Research Question:** Do the experimentally measured resonance frequencies align with the theoretical predictions for some resonance modes?

II. METHODS

A. Experimental Design

We present the theoretical model that we applied to a rectangular cavity. For an enclosed box cavity, we assumed the boundary conditions for the sound pressure P [3]:

$$\left.\frac{\partial P}{\partial x}\right|_{x=0} = \left.\frac{\partial P}{\partial x}\right|_{x=L_x} = \left.\frac{\partial P}{\partial y}\right|_{y=0} = \left.\frac{\partial P}{\partial y}\right|_{y=L_y} = \left.\frac{\partial P}{\partial z}\right|_{z=0} = \left.\frac{\partial P}{\partial z}\right|_{z=L_z} = 0$$

Where L_x, L_y, L_z represents length, width, and height. Then consider solving the wave equation:

$$\nabla^2 P = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} + \frac{\partial^2 P}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 P}{\partial t^2},$$

by using the separation of variables, which led to the theoretical resonance frequency model [3]:

$$f_{n_x n_y n_z} = \frac{\omega_{n_x n_y n_z}}{2\pi} = \frac{v}{2}\sqrt{\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2}},$$

where $f_{n_x n_y n_z}$ represent the theoretical resonance frequency, and n_x, n_y, n_z are the number of antinodes in the length, width, and height directions of the cavity.

Using a ruler, first we measured the height, width, and length (L_x, L_y, L_z) of the wooden cavity. A microphone was positioned close to the midpoint of the cavity's bottom edge, and a Bluetooth speaker was positioned right in the middle at the bottom. Then we closed the cavity to approximate the assumed boundary conditions. We then connected the microphone to an external oscilloscope via an amplifier using a cable, and the Bluetooth speaker was used to play the drive signal. Figure 1 shows all the equipment required for this experiment and their arrangement in the $z - y$ cross-section. Figure 2 on the following page provides a 3-D representation of the exact positions of the microphone and speaker within the rectangular cavity.

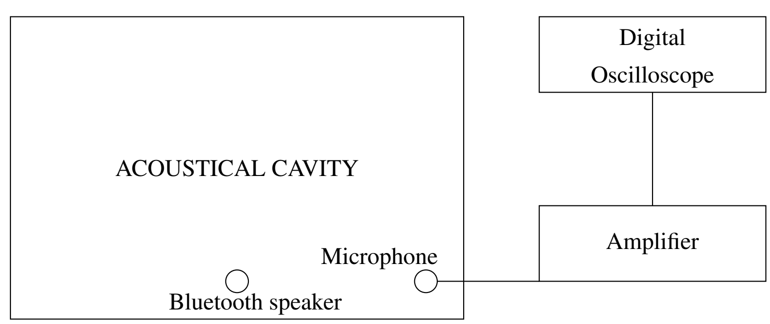

FIG. 1. Basic diagram of the experimental setup (cross-sectional view in the z-y plane)

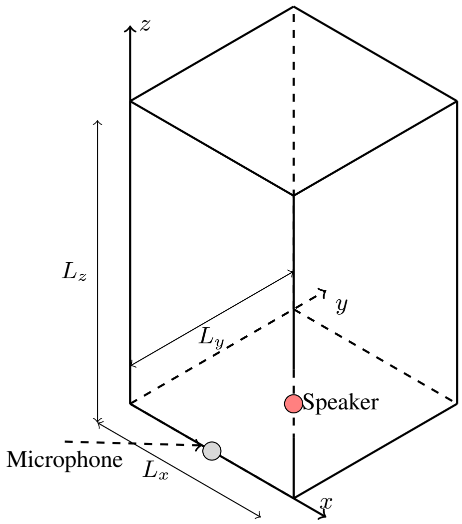

FIG. 2. Detailed positions of microphone and speaker within the rectangular cavity (3-D view)

When a significant increase followed by a decrease in amplitude was observed on the oscilloscope, the corresponding frequency was identified as a resonance frequency. Next, we moved the microphone from the bottom to the top of the cavity while maintaining the same driven frequency. By analyzing the amplitude changes on the oscilloscope, we determined the number of anti-nodes in the z-direction (n_z). Then, we found the set of resonance modes (n_x, n_y) on the prediction table, where the prediction value was closest to the measured resonance frequency, so that we determined nx, ny, nz and the corresponding predicted values. This procedure was repeated for all significant resonance peaks within the $0-800Hz$ range. Notice that the step size for frequency measurements was $1Hz$. Initially, when increasing the driven frequency, we used a larger step size $20Hz$ to identify possible resonance peaks. Once a peak was identified, we then verified its exact position using a smaller step size of $1Hz$ around the initial estimation. Our aim was to clearly identify all significant resonance peaks within the measured frequency range, ignoring minor or overlapping peaks.

B. Uncertainty Analysis

We consider that there are several main sources of uncertainty in the measured values: environmental noise interference, the step size of the drive frequency, and fluctuations on the oscilloscope. On the other hand, the main sources of uncertainty in the theoretically predicted values are the error of the ruler when measuring the geometric characteristics of the rectangular cavity and the non-absolute rigidity of the boundary conditions (sound pressure is absorbed or transmitted through wooden walls such that the partial derivative of sound pressure on the wall is not strictly 0). In summary, the uncertainty of the measured values is caused by external random errors and is maintained between 3 and $5Hz$. The uncertainty of the predicted values is mainly caused by the accuracy of the ruler, because when we slightly adjust the uncertainty of the side length, it may cause a fluctuation of $10^{+}Hz$, which is the biggest factor affecting the accuracy of the test results. In addition, for the error caused by the non-absolute rigidity of the walls, we need to consider more complex models in the future.

C. Result Test

We have quantified the uncertainty by using a 68% confidence interval (standard deviation σ), so its probability density function is $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, where μ is the mean or expectation of the distribution [4]. To quantitatively evaluate the agreement between measured and predicted resonance frequencies, we define the overlap ratio based on their respective Gaussian probability density functions:

- **Overlap Area** (shared area between two distributions):

$$\int_a^b \min\left(f_{\text{measured}}(x), f_{\text{predicted}}(x)\right) dx.$$

- **Measured Area** (normalization factor for the measured distribution):

$$\int_a^b f_{\text{measured}}(x) dx.$$

- **Overlap Ratio**:

$$\frac{\int_a^b \min\left(f_{\text{measured}}(x), f_{\text{predicted}}(x)\right) dx}{\int_a^b f_{\text{measured}}(x) dx},$$

  Here, the interval $[a, b]$ represents the frequency range of interest, which includes the entire peak of distributions.

We regard the overlap ratio as a quantitative indicator of the agreement between the measured and predicted frequencies and classify the values as follows:

- **Good agreement**: $75\% \leq OverlapRatio \leq 100\%$.
- **Reasonable agreement**: $50\% \leq OverlapRatio \leq 74\%$.
- **Limited agreement**: $25\% \leq OverlapRatio \leq 49\%$.
- **Poor agreement**: $0\% \leq OverlapRatio \leq 24\%$.

This classification allows the predictive performance of the models to be systematically evaluated based on a comparison of the probability of resonance frequencies.

III. RESULTS AND DISCUSSION

We recorded all significant resonance frequencies within the 0-800 Hz range and identified the corresponding most

TABLE I. Comparison of predicted and measured resonance frequencies with uncertainties.

Resonance Mode (n_x, n_y, n_z)	**Prediction Frequency (Hz)**	**Uncertainty_P (Hz)**	**Measured Frequency (Hz)**	**Uncertainty_M (Hz)**
$[0, 0, 1]$	143.3	5.97	147	3.5
$[0, 0, 2]$	286.7	11.94	308	4.5
$[0, 0, 3]$	430.0	17.92	444	4.5
$[0, 0, 4]$	573.3	23.89	588	4.5
$[0, 0, 5]$	716.7	29.86	735	4.0
$[0, 2, 1]$	792.2	17.39	799	5.0
$[2, 0, 1]$	792.2	17.39	799	5.0

likely resonance mode (n_x, n_y, n_z) for each peak and calculated the theoretical resonance frequency. These results are summarized in Table 1.

We also recorded the geometric properties of the rectangular, the temperature on the day of the experiment, and the assumed speed of sound to make it easy for readers to reproduce our experimental results.

TABLE II. Geometric properties of the cavity and assumed speed of sound.

Quantity	**Value**	**Uncertainty**
Length (L_x)	$0.44m$	$0.01m$
Width (L_y)	$0.44m$	$0.01m$
Height (L_z)	$1.20m$	$0.05m$
Room Temperature (T_0)	$294.55K$	$0.06K$
Speed of Sound (v_{ideal})	$344m/s$	–

As shown in Table I, there is a general agreement between the measured resonance frequencies and the theoretical predictions; a more reliable and quantitative assessment requires the use of the Gaussian overlap method introduced earlier. Here, we would like to discuss some of the findings from the experiment.

To identify the number of antinodes along the z-axis (n_z), we moved the microphone vertically from the bottom to the top of the cavity while maintaining a fixed driving frequency. This procedure is based on the assumption that, when the microphone is placed at the center of the bottom edge of the cavity, it should be located at an antinode along the z-axis, where the sound pressure amplitude is expected to be maximal. As the microphone was raised along the z-axis, we observed periodic variations in amplitude, with alternating peaks and troughs. This periodicity is a strong indication of the stationary wave structure in the vertical direction and provides a reliable method to count the number of antinodes and thus determine n_z. Despite the potential influence of horizontal mode mixing (i.e., nonzero n_x and n_y components), which can cause spatial pressure variations that deviate from the ideal 1D case, the vertical amplitude profile remained sufficiently regular to allow reliable identification of n_z.

Another point worth noting is that, in the lower frequency range shown in Table I, the identified resonance modes tend to zero antinodes in the x and y-directions ($n_x = n_y = 0$). We believe this is due to the fact: in this lower frequency range, resonance peaks associated with more complex spatial modes may have been obscured or overlapped by stronger, more dominant peaks corresponding to simpler modes. This overlap effect can lead to some modes being underrepresented or entirely missed in the experimental data.

Figure 3 on the following page shows the Gaussian probability density functions for all measured and predicted resonance frequencies. Based on the computed overlap ratios, which from lowest to highest frequency are 63%, 17%, 33%, 29%, 22%, 47%, and 47%, we observe that only the first mode achieves a level of agreement as "reasonable." All remaining modes fall into the "limited" or "poor" agreement.

Discussion: We analyzed the reasons for the above findings. On the one hand, we notice a consistent trend in which the predicted frequencies are systematically lower than the measured ones. This systematic underestimate indicates that, even though the identification of resonance modes (particularly n_z) is reliable, the theoretical model does not capture the behavior of the rectangular cavity. This discrepancy is probably due to two primary factors:

- (i) Overlapping of resonance peaks at the $0 - 800Hz$ frequency range, which makes it difficult to isolate individual modes.
- (ii) The effect of a wooden cavity wall, which may vibrate or leak sound, thus contradicting the assumption of a perfectly rigid boundary. Therefore, we provide here a mathematical proof of why walls that are not absolutely rigid can lead to predicted values that are lower than the actual values: When $\frac{\partial P}{\partial n}\big|_{\text{boundary}} = 0$, we can assume $P(x) = A\cos(kx) + B\sin(kx)$ in one dimension, and then the stationary wave satisfies $kL = n\pi \to f_n = \frac{vn}{2L}$. When $\frac{\partial P}{\partial n}\big|_{\text{boundary}} = -\alpha P \neq 0 \quad (\alpha > 0)$, the resonance condition can becomes approximately $\tan(kL) = \frac{\alpha}{k} > 0$, thus if we compared to the theoretical model, we can see that the actual situation will produce an overall frequency shift.

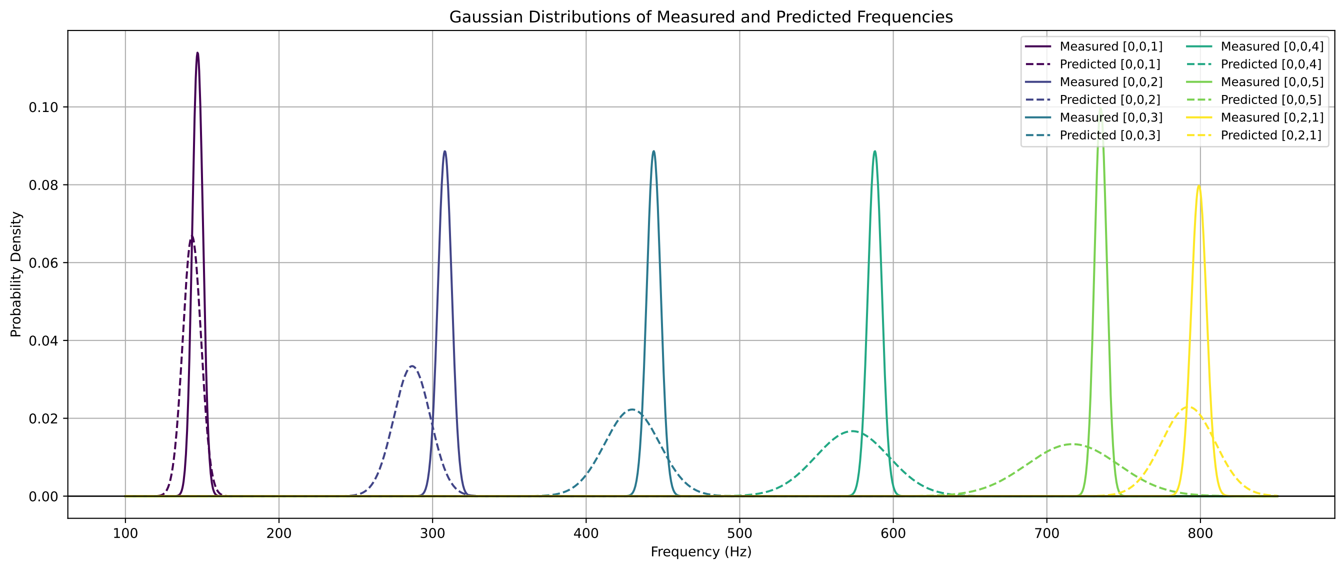

FIG. 3. Gaussian Distributions of the values of Table I

IV. CONCLUSIONS

This study was set out to analyze the acoustic resonance behavior of a rectangular wooden cavity by comparing experimentally measured resonance frequencies with theoretically predicted values. This study has found that, while the measured resonance frequencies generally follow the theoretical trend, systematic discrepancies exist between the predicted and experimental values. The results of this experiment demonstrate that a rectangular wooden cavity with dimensions 0.44 ± 0.01 m in length and width, and 1.20 ± 0.05 m in height, shows six significant cavity natural resonance frequencies at 147 ± 3.5 Hz, 308 ± 4.5 Hz, 444 ± 4.5 Hz, 588 ± 4.5 Hz, 735 ± 4.0 Hz, and 799 ± 5.0 Hz when driven by sound in the frequency range of $0 - 800Hz$ with an assumed speed of sound of 344 m/s. In this case, the theoretical model produce an overlap ratio of only 63%, 17%, 33%, 29%, 22%, and 47% respectively, all of which showed a consistent tendency to underestimate the actual measured values. The final analysis concluded that this systematic underestimation arises from the nonideal rigidity of the cavity walls, which was confirmed both through physical observation of wall vibrations and sound leakage and through mathematical analysis involving boundary conditions. The principal theoretical implication of this study is that the application of an idealized model with completely rigid boundaries to real acoustic systems can lead to significant differences, especially in enclosed cavities made of wood or flexible materials. At the same time, we have found an effective method to identify theoretical resonance modes and resonance characteristics in the cavity. In addition, we have created the "overlap rate" method, which provides a way to evaluate the degree of agreement between measured and theoretical values for resonance systems. The limitations of this study are that the oscilloscope cannot identify important peaks that are overlapped, which makes it impossible to fully capture the complexity of the three dimensional pattern and its interactions in the cavity. It has also been mentioned repeatedly above that we have not taken into consideration theoretical models under non-rigid boundaries, which leads to systematic deviations in the prediction. Future research should consider attempting to model a complex theoretical model with a non-rigid boundary, analyze the resonance characteristics in the cavity after effectively identifying possible resonance frequencies, and attempt to visualize such resonance characteristics to thoroughly study the resonance in the cavity.

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